THEODORA MOLDOVAN

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RESEARCH INTERESTS

I work at the intersection of network science, sociology, and optimization, developing computational methods to understand network inequalities and to design algorithms that promote fairness in social systems.

EDUCATION

PhD in Computer Science expected January 2031
Uppsala University, Uppsala, Sweden
Thesis: Fair Social Network Analysis

MSc in Data Science June 2025
Uppsala University, Uppsala, Sweden
Specialization: Machine Learning and Statistics
Thesis: Exploring Collective Action in Black Lives Matter Podcast Conversations

BA in Statistics and Data Science May 2023
Connecticut College, New London, Connecticut, USA
Minor: Computer Science
Thesis: Moderated Parallel Mediation Analysis with Applications on the Qualitative Job Insecurity Scale

ONGOING WORK

Moldovan, T., Magnani, M., Flener, P., & Bask, M. (2026). Social stratification in a population-scale social network.

Adel, G., Bhuria, S., Catalano, A., Dorizzi, L., Federici, L., **Moldovan, T.**, Nortier, B., Punzal, C.D., de Meijere, G., & Iñiguez, G. (2026). Preferentiality and bandwidth drive tie activity in online and offline ego networks. `https://arxiv.org/abs/2606.27937`

Moldovan, T., Pera, A., Vega, D., & Aiello, L. M. (2025). Podcasts as a Medium for Participation in Collective Action: A Case Study of Black Lives Matter. `https://arxiv.org/abs/2509.13197`

PUBLICATIONS

Moldovan, T., Panayiotou, G., & Vega, D. (2026). Les Misérables Dialogue: A film-based network representation of social communication. *International Conference on Complex Networks and Their Applications*. In press.

David Avritzer, J., Zhuang, Y., Avilez Lopez, M., **Moldovan, T.**, & Nguyen, N. (2025). Autonomous Demand and Its Impact on Growth and Income Distribution: Empirical Insights from South America. *Review of Radical Political Economics*. `https://doi.org/10.1177/04866134261464758`.

Abu Khait, A., Menger, A., Rababa, M., **Moldovan, T.**, Lazenby, M., & Shellman, J. (2024). The Mediating Role of Religion and Loneliness on the Association between Reminiscence Functions and Depression: A Call to Advance Older Adults' Mental Health. *Psychogeriatrics*. `https://doi.org/10.1111/psyg.13041`

Abu Khait, A., Menger, A., Al-Modallal, H., Abdalrahim, A., **Moldovan, T.**, & Hamaideh, S.H. (2024). Self-Transcendence as a Mediator of the Relationship Between Reminiscence Functions and Death Anxiety: Implications for Psychiatric Nurses. *Journal of the American Psychiatric Nurses Association*. `https://doi.org/10.1177/10783903231174464`

PRESENTATIONS

Social stratification in a population-scale social network September 2026
The 1st Nordic Conference on Computational Social Science (CS2Nordics)
Podcasts as a medium for participation in collective action September 2026
Statistical Research course, Department of Mathematics and Statistics, Connecticut College, USA
Fairness in social network analysis September 2026
Data, Ethics and Law course, Department of Information Technology, Uppsala University, Sweden
Individual fairness and social network analysis April 2026
Seminar in Mathematics, Department of Mathematics and Statistics, Connecticut College, USA
Podcasts as a medium for participation in collective action November 2025
American Statistical Association, Detroit Chapter, USA

SELECTED RESEARCH EXPERIENCE

PhD Candidate February 2026 - present
InfoLab, Department of Information Technology, Uppsala University, Sweden
Research Assistant September 2025 - January 2026
Urban Lab and Uppsala Immigration Lab, Uppsala University, Sweden
Research Assistant April 2025 - June 2025
Department of Economics, Stockholm University, Sweden
Research Assistant November 2024 - June 2025
Department of Peace and Conflict Research, Uppsala University, Sweden
Research Assistant June 2024 - December 2024
Institute for Futures Studies, Sweden

Research Assistant January 2024 - June 2024
Department of Statistics, Uppsala University, Sweden
Data Analytics Intern June 2022 - August 2022
Analytics Center of Excellence, Yale School of Medicine, Yale University, USA
Statistical Research Intern January 2022 - May 2022
Menger Analytics, USA

TEACHING EXPERIENCE

Teaching Assistant September 2023 - present
Department of Information Technology, Uppsala University, Sweden

- Visual Communication of Data (1DL415) Fall 2026
- Introduction to Machine Learning (1DL034) Spring 2025
- Data, Ethics and Law (1DL002) Fall 2024, Fall 2026
- Algorithms and Data Structures I (1DL210) Fall 2023, Fall 2024
- Program Design and Data Structures (1DL201) Fall 2023, Spring 2024
- Data Mining (1DL360) Fall 2023, Fall 2026

Teaching Assistant September 2020 - May 2023
Department of Computer Science & Department of Mathematics and Statistics, Connecticut College, USA

- Biological Statistics (BIO 235/STA 235) Spring 2023
- Introduction to Statistics (STA 107) Spring 2022
- Machine Learning and Data Mining (COM 307) Fall 2021, Fall 2022
- Advanced Regression Techniques (STA 207) Spring 2021, Fall 2021, Fall 2022
- Introduction to Computer Science and Problem Solving (COM 110) Fall 2020, Spring 2021, Fall 2021, Spring 2022

SUPERVISION

Simen Odden Hansen
MSc Thesis in Data Science, Uppsala University, Sweden Spring 2027
Anton Eckervald & Samuel Westman
Project in Data Science, Uppsala University, Sweden Fall 2026

LEADERSHIP

Program Assistant August 2024 - August 2025
Master's Program in Data Science, Uppsala University, Sweden
Social Media Coordinator November 2021 - May 2023
Ammerman Center for Arts and Technology, Connecticut College, USA
President, Vice President August 2021 - September 2022
International Student Association, Connecticut College, USA
International Adviser & Coordinator August 2020 - December 2021
Office of International Students, Connecticut College, USA
Resident Assistant August 2021 - May 2023
Residential Education and Living, Connecticut College, USA

DISTINCTIONS AND AWARDS

Distinction in the Major Field May 2023
Department of Mathematics and Statistics, Connecticut College, USA
Senior Julia Wells Bower Prize April 2023
Department of Mathematics and Statistics, Connecticut College, USA
Smalley/Zahler Award April 2023
Ammerman Center for Arts and Technology, Connecticut College, USA
Winthrop Scholar February 2023
Phi Beta Kappa, Delta of Connecticut Chapter, Connecticut College, USA
Presidential Scholar August 2019
Founders Scholarship (USD 100,000), Connecticut College, USA

ACADEMIC MEMBERSHIPS

Nordic Society for Computational Social Science (NoSoCSS) since February 2026
Uppsala University Network on Computational Social Science since February 2026
Phi Beta Kappa since May 2023
Pi Mu Epsilon since May 2023
American Statistical Association (ASA) since April 2022

REFERENCES

Matteo Magnani, Professor, Department of Information Technology, Uppsala University matteo.magnani@it.uu.se
Davide Vega, Associate Professor, Department of Information Technology, Uppsala University davide.vega@it.uu.se
Luca Maria Aiello, Professor, Data Science Section, IT University of Copenhagen luai@itu.dk